Coplanar Vectors

4. Write each of the following vectors as a linear combination of $\hat{i}$ and $\hat{j}$.

a. the vector $\vec{p} = (-4, 5)$

b. the position vector of the point $A(8, -3)$

c. a vector directed at an angle of 45° with a magnitude of $\sqrt{2}$

d. a vector directed at an angle of 150° with a magnitude of 6

5. a. Can every vector in the xy-plane be written as a linear combination of $\vec{u} = (1, 4)$ and $\vec{v} = (-2, 5)$? Justify your answer.

b. Write the vector $(-567, -669)$ in terms of $\vec{u}$ and $\vec{v}$.

6. Can every vector in the xy-plane be written as a linear combination of $\vec{u} = (-4, -6)$ and $\vec{v} = (10, 15)$? Justify your answer.

Part B

7. Are the following sets of vectors coplanar?

a. $(1, -1, 1), (0, 1, 1), (1, 0, 2)$ b. $(1, 0, 1), (1, 1, 1), (1, 0, -1)$

8. If $\vec{u} = (2, 1, 1)$ and $\vec{v} = (-1, 1, 3)$

a. which of the following vectors can be written in the form $s\vec{u} + t\vec{v}$

(i) $(4, 2, 2)$ (ii) $(1, 2, 4)$ (iii) $(1, 5, 11)$ (iv) $(4, 5, 8)$

b. Find another vector that can be expressed in the form $s\vec{u} + t\vec{v}$.

c. Find another vector which *cannot* be expressed in the form $s\vec{u} + t\vec{v}$, and explain why it cannot.

9. Given that

$$\vec{u} = x\vec{a} + 2y\vec{b}$$
$$\vec{v} = -2y\vec{a} + 3y\vec{b}$$
$$\vec{w} = 4\vec{a} - 2\vec{b}$$

where $\vec{a}$ and $\vec{b}$ are not collinear, find the values of x and y for which $2\vec{u} - \vec{v} = \vec{w}$.

Part C

10. Find values of a, b, and c which satisfy each of the following equations.

a. $a(2, 1, 0) + b(-3, 4, 5) + c(2, 0, 3) = (-4, 10, 7)$

b. $a(3, -1, 2) + b(-1, 1, 3) + c(2, 1, 5) = (2, 5, 16)$

11. By using the triple scalar product, determine if the following vectors are coplanar.

a) $(1, -1, 1), (0, 1, 1), (1, 0, 2)$ b) $(1, 0, 1), (1, 1, 1), (1, 0, -1)$

Answers

4. a. $\vec{p} = -4\hat{i} + 5\hat{j}$ **b.** $\overrightarrow{OA} = 8\hat{i} - 3\hat{j}$ **c.** $\hat{i} + \hat{j}$ **d.** $-3\sqrt{3}\hat{i} + 3\hat{j}$
5. a. yes **b.** $-321\vec{u} + 123\vec{v}$ **6.** No **7.a.** yes **b.** no **8. a.** parts (i), (ii), (iii) **9.** 4, −2 **10. a.** 2, 2, −1 **b.** −1, 1, 3 **11.** a) yes b) no